Hypothesis Testing
Special Case of Two Groups

I. Comparing Two Groups

- A. Idea
- B. t Test Procedure
- C. Example – Fasting Glucose
- D. Example – Halothane vs. Morphine

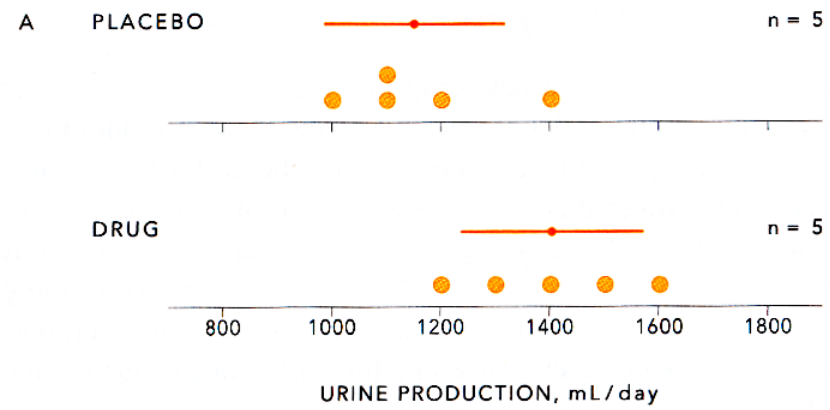
II. t Test for One-Sample Inference

Homework 1

I. Overview/explanation of problems
II. Help with questions on material

A. Idea

$n = 5$ case



$n = 20$ case

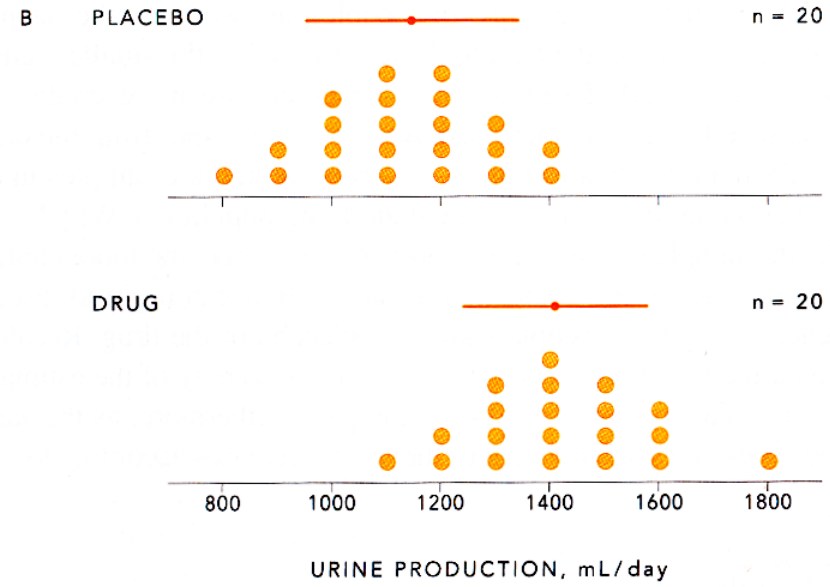
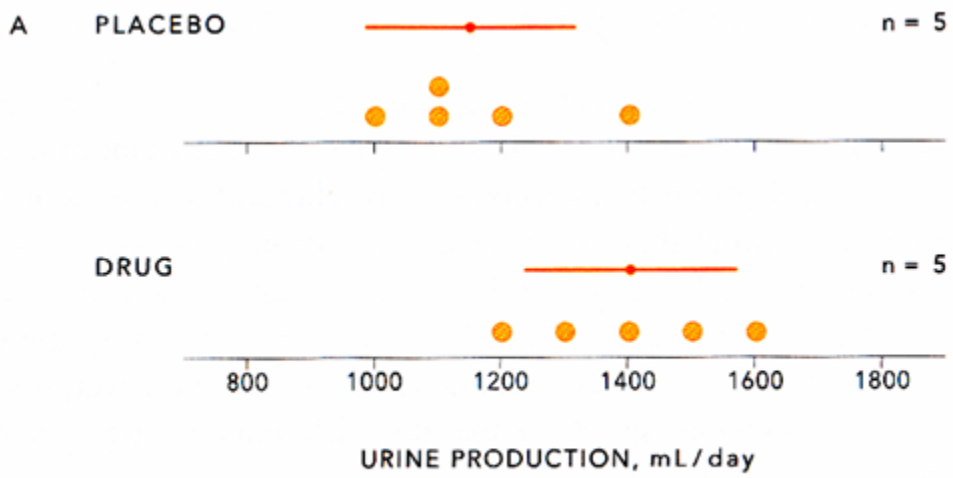


Figure 4–2 ***(A)*** Results of a study in which five people were treated with a placebo and five people were treated with a drug thought to increase daily urine production. On the average, the five people who received the drug produced more urine than the placebo group. Are these data convincing evidence that the drug is an effective diuretic? ***(B)*** Results of a similar study with 20 people in each treatment group. The means and standard deviations associated with the two groups are similar to the results in panel ***A***. Are these data convincing evidence that the drug is an effective diuretic? If you changed your mind, why did you do it?

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

Quantifies Uncertainty in Estimate of Pop. Mean



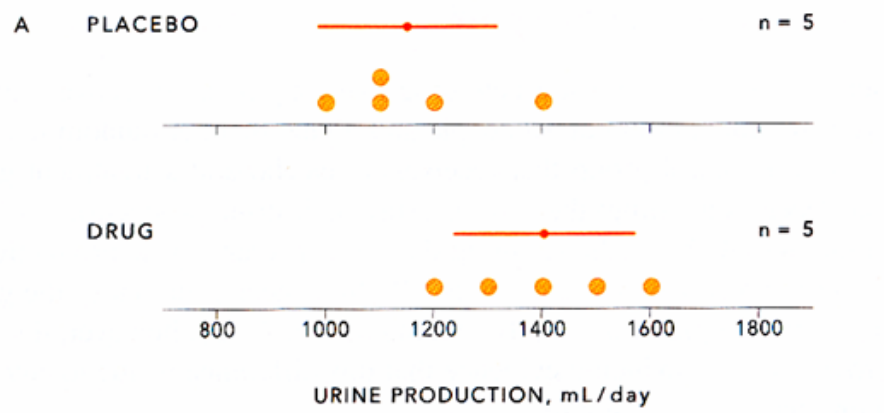
$$\frac{\text{difference in sample means}}{\text{standard error of difference of sample means}}$$

Define a new statistic: t

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_{(\bar{x}_1 - \bar{x}_2)}}$$

How to calculate?

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_{(\bar{x}_1 - \bar{x}_2)}}$$



From data: $\bar{x}_1, s_1^2, \bar{x}_2, s_2^2$

Also: $s_{\bar{x}_1} = \frac{s_1}{\sqrt{n_1}}, \; s_{\bar{x}_2} = \frac{s_2}{\sqrt{n_2}}$

But need: $s_{(\bar{x}_1 - \bar{x}_2)}$

Can show: $s^2_{(\bar{x}_1 - \bar{x}_2)} = s^2_{\bar{x}_1} + s^2_{\bar{x}_2}$

Recall from Probability

$$Z = X + Y \quad \text{or} \quad Z = X - Y \qquad X \sim f(x) \quad Y \sim f(y)$$

X and Y are independent:

$$E\left[XY \right] = \mu_X \mu_Y$$

$$E[h(X,Y)] = \int_{-\infty}^{\infty} \int h(x,y) f(x,y)\, dx\, dy = \int_{-\infty}^{\infty} \int h(x,y) f(x) f(y)\, dx\, dy$$

$$E[X+Y] = \int_{-\infty}^{\infty} \int x f(x) f(y)\, dx\, dy + \int_{-\infty}^{\infty} \int y f(x) f(y)\, dx\, dy$$

$$E[X+Y] = \int_{-\infty}^{\infty} x f(x) dx \cdot \int_{-\infty}^{\infty} f(y) dy + \int_{-\infty}^{\infty} y f(y) dy \cdot \int_{-\infty}^{\infty} f(x) dx$$

$$E[Z] = E[X \pm Y] = E[X] \pm E[Y] = \mu_X \pm \mu_Y \equiv \mu_Z$$

Recall from Probability

$$Z = X + Y \quad \text{or} \quad Z = X - Y \qquad X \sim f(x) \quad Y \sim f(y)$$

X and Y are independent:

$$E[XY] = \mu_X \mu_Y$$

But we want:

$$\sigma^2_{X-Y} = \sigma^2_Z \equiv E[(Z - \mu_Z)^2] = E[Z^2] - (\mu_Z)^2$$

$$= E[(X - Y)^2] - (\mu_X - \mu_Y)^2$$

$$\vdots$$

$$= \sigma^2_X + \sigma^2_Y$$

$\boxed{Z = X - Y}$

By same methodology: $\sigma^2_{X+Y} = \sigma^2_X + \sigma^2_Y$

Therefore

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_{(\bar{x}_1 - \bar{x}_2)}} = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_{\bar{x}_1}^2 + s_{\bar{x}_2}^2}} = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

<u>However, under the null hypothesis, the pooled variance is a better estimate for σ^2</u>

Replace $s_1{}^2$ and $s_2{}^2$ with $$s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

<u>Pooled Variance</u>

B. t Test Procedure (Student t Test)

Experiment:

RV X is measured — 2 treatment groups

n_1, n_2 # number of subjects in groups 1 & 2

Null Hypothesis:

$$H_0 : \mu_1 = \mu_2 \qquad H_1 : \mu_1 \neq \mu_2$$

1. Mean of each group

$$\bar{x}_1,\ \bar{x}_2 \qquad \left(\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i \right)$$

2. Variance of each group

$$s_1^2,\ s_2^2 \qquad \left(s_i^2 = \frac{1}{n-1} \sum_{i=1}^{n} \left(x_i - \bar{x} \right)^2 \right)$$

B. t Test Procedure (Student t Test)

3. Pooled Variance

$$s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

4. Compute Value of t

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s^2}{n_1} + \frac{s^2}{n_2}}} = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$\nu = n_1 + n_2 - 2$$

Degrees of freedom

B. t Test Procedure (Student t Test)

5. Determine t_{crit} from t table

$$t_{crit} \equiv t_{\alpha}(\nu) \qquad \alpha\text{: significance level}$$

Using a two-tailed t table

6. Compare t to t_{crit} and State Conclusion

If $|t| > t_{\alpha}(\nu)$ Reject null hypothesis with $P < \alpha$

Assumptions of t Test

1. Data are interval data
2. Samples must be independent
3. Samples must be randomly selected
4. Population(s) must be Normally distributed
5. Variance(s) of each population must be equal

If $|t| > t_{\alpha}(\nu)$

Reject null hypothesis: $P < \alpha$

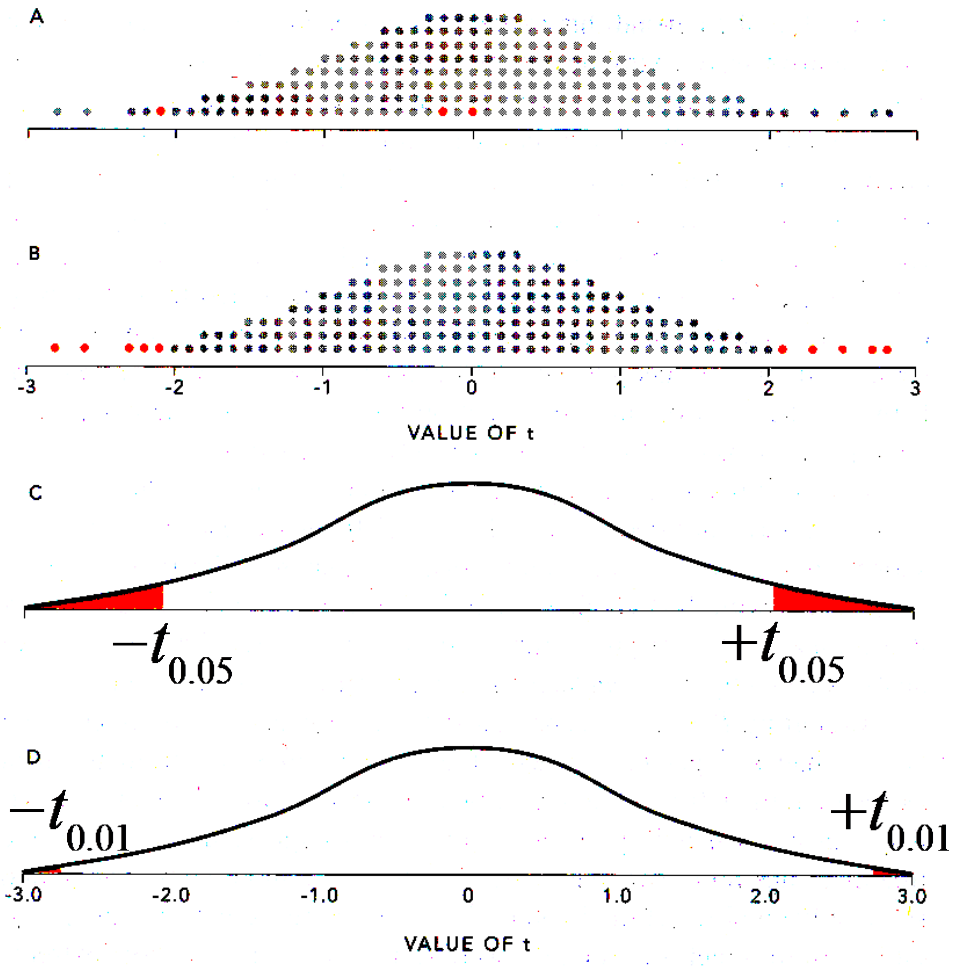


Figure 4–5 The results of 200 studies like that described in Fig. 4–4; the three specific studies from Fig. 4–4 are indicated in panel ***A***. Note that most values of the *t* statistic cluster around 0, but it is possible for some values of *t* to be quite large, exceeding 1.5 or 2. Panel ***B*** shows that there are only 5 chances in 100 of *t* exceeding 2.1 in magnitude if the two samples were drawn from the same population. If one continues examining all possible samples drawn from the population and our pairs of samples drawn from the same population, one obtains a distribution of all possible *t* values which becomes the smooth curve in panel ***C***. In this case, one defines the critical value of *t* by saying that it is unlikely that this value of *t* statistic was observed under the hypothesis that the drug had no effect by taking the 5 percent most extreme error areas under the tails of distribution and selecting the *t* value corresponding to the beginning of this region. Panel ***D*** shows that if one required a more stringent criterion for rejecting the hypothesis for no difference by requiring that *t* be in the most extreme 1 percent of all possible values, the cutoff value of *t* is 2.878.

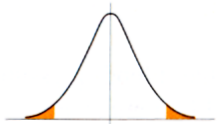


Table 4–1 Critical Values of *t* (Two-Tailed)

	Probability of greater value, *P*								
ν	0.50	0.20	0.10	0.05	0.02	0.01	0.005	0.002	0.001
1	1.000	3.078	6.314	12.706	31.821	63.657	127.321	318.309	636.619
2	0.816	1.886	2.920	4.303	6.965	9.925	14.089	22.327	31.599
3	0.765	1.638	2.353	3.182	4.541	5.841	7.453	10.215	12.924
4	0.741	1.533	2.132	2.776	3.747	4.604	5.598	7.173	8.610
5	0.727	1.476	2.015	2.571	3.365	4.032	4.773	5.893	6.869
6	0.718	1.440	1.943	2.447	3.143	3.707	4.317	5.208	5.959
7	0.711	1.415	1.895	2.365	2.998	3.449	4.029	4.785	5.408
8	0.706	1.397	1.860	2.306	2.896	3.355	3.833	4.501	5.041
9	0.703	1.383	1.833	2.262	2.821	3.250	3.690	4.297	4.781
10	0.700	1.372	1.812	2.228	2.764	3.169	3.581	4.144	4.587
11	0.697	1.363	1.796	2.201	2.718	3.106	3.497	4.025	4.437
12	0.695	1.356	1.782	2.179	2.681	3.055	3.428	3.930	4.318
13	0.694	1.350	1.771	2.160	2.650	3.012	3.372	3.852	4.221
14	0.692	1.345	1.761	2.145	2.624	2.977	3.326	3.787	4.140
15	0.691	1.341	1.753	2.131	2.602	2.947	3.286	3.733	4.073
16	0.690	1.337	1.746	2.120	2.583	2.921	3.252	3.686	4.015
17	0.689	1.333	1.740	2.110	2.567	2.898	3.222	3.646	3.965
18	0.688	1.330	1.734	2.101	2.552	2.878	3.197	3.610	3.922
19	0.688	1.328	1.729	2.093	2.539	2.861	3.174	3.579	3.883
20	0.687	1.325	1.725	2.086	2.528	2.845	3.153	3.552	3.850
21	0.686	1.323	1.721	2.080	2.518	2.831	3.135	3.527	3.819
22	0.686	1.321	1.717	2.074	2.508	2.819	3.119	3.505	3.792
23	0.685	1.319	1.714	2.069	2.500	2.807	3.104	3.485	3.768
24	0.685	1.318	1.711	2.064	2.492	2.797	3.091	3.467	3.745
25	0.684	1.316	1.708	2.060	2.485	2.787	3.078	3.450	3.725
26	0.684	1.315	1.706	2.056	2.479	2.779	3.067	3.435	3.707
27	0.684	1.314	1.703	2.052	2.473	2.771	3.057	3.421	3.690
28	0.683	1.313	1.701	2.048	2.467	2.763	3.047	3.408	3.674
29	0.683	1.311	1.699	2.045	2.462	2.756	3.038	3.396	3.659
30	0.683	1.310	1.697	2.042	2.457	2.750	3.030	3.385	3.646
31	0.682	1.309	1.696	2.040	2.453	2.744	3.022	3.375	3.633
32	0.682	1.309	1.694	2.037	2.449	2.738	3.015	3.365	3.622
33	0.682	1.308	1.692	2.035	2.445	2.733	3.008	3.356	3.611
34	0.682	1.307	1.691	2.032	2.441	2.728	3.002	3.348	3.601
35	0.682	1.306	1.690	2.030	2.438	2.724	2.996	3.340	3.591

(*continued*)

Table 4–1 Critical Values of *t* (Two-Tailed) (*Continued*)

	Probability of greater value, *P*								
ν	0.50	0.20	0.10	0.05	0.02	0.01	0.005	0.002	0.001
36	0.681	1.306	1.688	2.028	2.434	2.719	2.990	3.333	3.582
37	0.681	1.305	1.687	2.026	2.431	2.715	2.985	3.326	3.574
38	0.681	1.304	1.686	2.024	2.429	2.712	2.980	3.319	3.566
39	0.681	1.304	1.685	2.023	2.426	2.708	2.976	3.313	3.558
40	0.681	1.303	1.684	2.021	2.423	2.704	2.971	3.307	3.551
42	0.680	1.302	1.682	2.018	2.418	2.698	2.963	3.296	3.538
44	0.680	1.301	1.680	2.015	2.414	2.692	2.956	3.286	3.526
46	0.680	1.300	1.679	2.013	2.410	2.687	2.949	3.277	3.515
48	0.680	1.299	1.677	2.011	2.407	2.682	2.943	3.269	3.505
50	0.679	1.299	1.676	2.009	2.403	2.678	2.937	2.261	3.496
52	0.679	1.298	1.675	2.007	2.400	2.674	2.932	3.255	3.488
54	0.679	1.297	1.674	2.005	2.397	2.670	2.927	3.248	3.480
56	0.679	1.297	1.673	2.003	2.395	2.667	2.923	3.242	3.473
58	0.679	1.296	1.672	2.002	2.392	2.663	2.918	3.237	3.466
60	0.679	1.296	1.671	2.000	2.390	2.660	2.915	3.232	3.460
62	0.678	1.295	1.670	1.999	2.388	2.657	2.911	3.227	3.454
64	0.678	1.295	1.669	1.998	2.386	2.655	2.908	3.223	3.449
66	0.678	1.295	1.668	1.997	2.384	2.652	2.904	3.218	3.444
68	0.678	1.294	1.668	1.995	2.382	2.650	2.902	3.214	3.439
70	0.678	1.294	1.667	1.994	2.381	2.648	2.899	3.211	3.435
72	0.678	1.293	1.666	1.993	2.379	2.646	2.896	3.207	3.431
74	0.678	1.293	1.666	1.993	2.378	2.644	2.894	3.204	3.427
76	0.678	1.293	1.665	1.992	2.376	2.642	2.891	3.201	3.423
78	0.678	1.292	1.665	1.991	2.375	2.640	2.889	3.198	3.420
80	0.678	1.292	1.664	1.990	2.374	2.639	2.887	3.195	3.416
90	0.677	1.291	1.662	1.987	2.368	2.632	2.878	3.183	3.402
100	0.677	1.290	1.660	1.984	2.364	2.626	2.871	3.174	3.390
120	0.677	1.289	1.658	1.980	2.358	2.617	2.860	3.160	3.373
140	0.676	1.288	1.656	1.977	2.353	2.611	2.852	3.149	3.361
160	0.676	1.287	1.654	1.975	2.350	2.607	2.846	3.142	3.352
180	0.676	1.286	1.653	1.973	2.347	2.603	2.842	3.136	3.345
200	0.676	1.286	1.653	1.972	2.345	2.601	2.839	3.131	3.340
∞	0.6745	1.2816	1.6449	1.9600	2.3263	2.5758	2.8070	3.0902	3.2905
Normal	0.6745	1.2816	1.6449	1.9600	2.3263	2.5758	2.8070	3.0902	3.2905

Source: Adapted from J. H. Zar, *Biostatistical Analysis* (2 ed.), Prentice-Hall, Englewood Cliffs, N.J., 1984, pp. 484–485, table B.3. Used by permission.

VOLUME VI MARCH, 1908 No. 1

BIOMETRIKA.

THE PROBABLE ERROR OF A MEAN.

BY STUDENT.





In 1899, William Sealy Gosset, a 23-year-old chemistry graduate, took up a job as brewer at Arthur Guinness, Son & Co, Ltd, in Dublin, Ireland. His task was to apply scientific methods to beer processing.

William Sealy Gosset

C. Example - Fasting Glucose

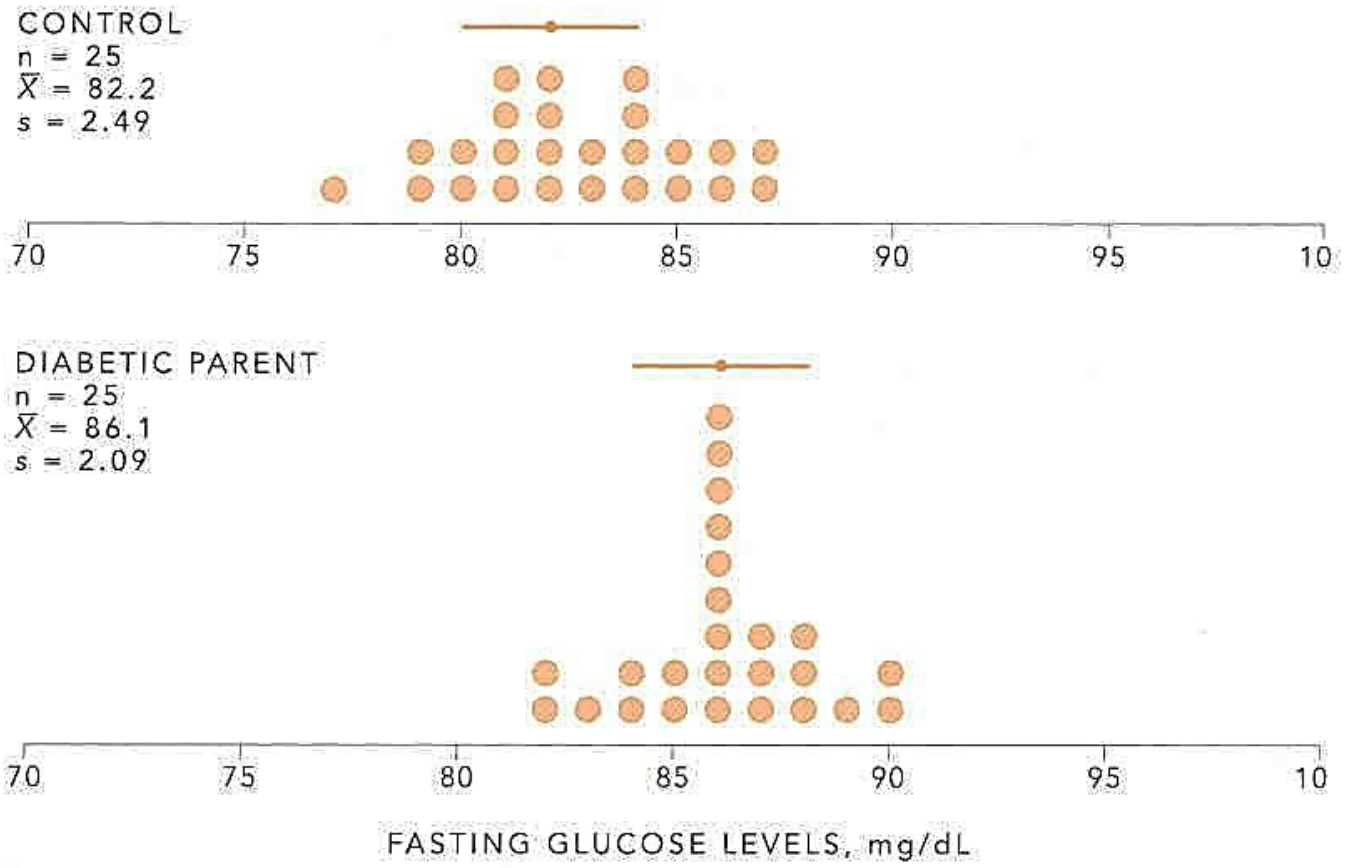


Figure 3–7 Results of a study comparing fasting glucose levels in children of parents who have type II diabetes and children of parents without type II diabetes. Each child's fasting glucose level is indicated by a circle at the appropriate glucose level. The average fasting glucose level for children of diabetic parents is higher than the average glucose level in children whose parents did not have diabetes. The statistical question is to assess whether or not the difference is due simply to random sampling or to an actual effect of the differences in parental history.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s^2}{n_1} + \frac{s^2}{n_2}}} \qquad s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$\bar{x}_1 = 82.2 \quad s_1 = 2.49 \qquad \bar{x}_2 = 86.1 \quad s_2 = 2.09$$

$$n_1 = 25, \; n_2 = 25$$

$$s^2 = \frac{1}{2}\left(s_1^2 + s_2^2\right) = \frac{1}{2}\left(2.49^2 + 2.09^2\right) = 5.28$$

$$t = \frac{82.2 - 86.1}{\sqrt{\left(5.28/25\right) + \left(5.28/25\right)}} = -6.001 \qquad \nu = n_1 + n_2 - 2 = 48$$

$$t_{0.05}(\nu = 48) = 2.011 \qquad t_{0.001}(\nu = 48) = 3.505$$

$$|t| = 6.001 > t_{0.001}(\nu = 48) = 3.505$$ Reject null hypothesis: $P < 0.001$

D. Example – Halothane vs. Morphine

Table 4–2 Comparison of Anesthetic Effects on the Cardiovascular System

	Halothane ($n = 9$)		Morphine ($n = 16$)	
	Mean	SD	Mean	SD
Best cardiac index, induction to bypass, $L/m^2 \cdot min$	2.08	1.05	1.75	.88
Mean arterial blood pressure at time of best cardiac index, mmHg	76.8	13.8	91.4	19.6
Total peripheral resistance associated with best cardiac index, $dyn \cdot s/cm^5$	2210	1200	2830	1130

Source: Adapted from T. J. Conahan et al., "A Prospective Random Comparison of Halothane and Morphine for Open-Heart Anesthesia," *Anesthesiology*, **38**:528–535, 1973.

	Halothane ($n = 9$)		Morphine ($n = 16$)	
	Mean	SD	Mean	SD
Best cardiac index induction to bypass, L/m² · min	2.08	1.05	1.75	.88

$$s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \quad ?????$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s^2}{n_1} + \frac{s^2}{n_2}}} = \quad ?????$$

$$\nu = n_1 + n_2 - 2 = \quad ?????$$

$$t_{0.05}(\nu) = \quad ?????$$

$|t| =$? $\left(> \text{or} \not> \right) t_{0.05}(\nu = \ ? \ ) = ?$? null hypothesis: $P <$?

	Halothane ($n = 9$)		Morphine ($n = 16$)	
	Mean	SD	Mean	SD
Best cardiac index induction to bypass, L/m² · min	2.08	1.05	1.75	.88

$$s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \quad 0.889$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s^2}{n_1} + \frac{s^2}{n_2}}} = \quad 0.840 \qquad \nu = n_1 + n_2 - 2 = \quad 23$$

$$t_{0.05}(\nu) = \quad 2.069$$

$$|t| = \quad 0.840 \quad \not> t_{0.05}(\nu = \ 23\ ) = 2.069$$

Do not reject null hypothesis;
Conclude no diff. in CI

II. t Test for One-Sample Inference

Experiment:

RV X is measured n : number of subjects

Null Hypothesis:

$$H_0 : \mu = \mu_0 \ \left(\mu - \mu_0 = 0\right) \qquad H_1 : \mu \neq \mu_0$$

1. Mean of group

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

2. Variance of group

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(x_i - \overline{x}\right)^2$$

3. Compute Value of t

Sample mean

Reference value against which you are comparing sample group

$$t = \frac{\bar{x} - \mu_0}{\sqrt{s^2 / n}}$$

$$\nu = n - 1$$

Degrees of freedom

4. Determine t_{crit} from two-tailed t table

$$t_{crit} \equiv t_{\alpha}(\nu)$$

α: significance level

5. Compare t and State Conclusion

If $|t| > t_{\alpha}(\nu)$ Reject null hypothesis with $P < \alpha$

Example: HR in BME 423

Experiment (Observations):

Measure HR in a sample of BME 423 students

$\bar{x} = 76.8 \qquad s^2 = (4.15)^2 \qquad n = 5$

Test the Null Hypothesis:

H_0: $\mu_{\mathrm{HR,423}} = 72$ $\qquad\qquad$ H_1: $\mu_{\mathrm{HR,423}} \neq 72$

units:
BPM for means
$(\mathrm{BPM})^2$ for variances

H0: There is no difference between the mean heart rate of BME 423 students and 72 BPM

$$t = \frac{\overline{x} - \mu_0}{\sqrt{s^2/n}} = \frac{76.8 - 72.0}{\sqrt{(4.15)^2/5}}$$

$$= \frac{4.8}{1.855936} = 2.586$$

$$\nu = n - 1 = 5 - 1 = 4$$

Table 4–1 Critical Values of t (Tw

	Probability of great				
ν	0.50	0.20	0.10	0.05	0.02
1	1.000	3.078	6.314	12.706	31.821
2	0.816	1.886	2.920	4.303	6.965
3	0.765	1.638	2.353	3.182	4.541
4	0.741	1.533	2.132	2.776	3.747
5	0.727	1.476	2.015	2.571	3.365

$$t_{crit} = t_{0.05}(\nu = 4) = 2.776$$

$$|t| = 2.586 \not> t_{crit} = 2.776$$

$\Rightarrow$ do not reject Ho (because $P > \alpha$)

conclusion: $\mu_{HR,423}$ is not significantly different from 72 BPM